

AnyCam 2.6.7 Audit Report

Release audit - 02 October 2026 - night boost, insects, infrared switches, file names - tag 2.6.7

2.6.7 adds build plan C15: under infrared, each armed camera's motion sensitivity rises by 15 on its own, and a sunrise and sunset check reports cameras that do not switch. Scope confirmed by CrystalHeeler on 2026-10-01: version 2.6.7 (SFTP and FTP moved to 2.6.8), +15 for every camera, no 'small changes must last' rule, night watched all the time, location from Home Assistant, missed switches reported three ways. A second round, the same day, after CrystalHeeler's overnight test on 2.6.6: insects and infrared-colour switches stop recording, and recording names start with the camera (CrystalHeeler: version 2.6.7, name choice A). Against 2.6.6: camera_discovery.py 392 lines added, 33 removed; verify_release.py 18 added, 3 removed.

Evidence for the change

Measurement	Value	Source
Distant approach on ch4, before reaching 1.1%	0.6-0.7% for 6-8 s	CrystalHeeler's log, 2026-10-01, 22:47:01-:07 and 22:49:33-:41
Still night scene, minute peaks	median 0.1%, highest 0.7%	Same log, 22:27-23:03
Most sensitive setting before 2.6.7	100 = 1.0%	2.6.6 scale
Night setting 80 + 15 = 95	1.24%	2.6.7 scale
Night setting 100 + 15 = 115	0.52%	2.6.7 scale

Floor 0.4% Never reached by 115; a safety bound
The 0.7% still-scene maximum is above 0.52%, so a camera at 100 by day can record a false event at night. CrystalHeeler chose +15 for every camera knowing this; each camera's slider still sets its own night level.

Design

Part	Design
Night signal	The detector's ffmpeg output changed from grey to YUV 4:2:0 (4,608 bytes per 64 x 48 frame). Once a second the colour planes give a measure: mean distance from neutral. At or under 2.5 for 30 s = night; at or over 5.0 for 30 s = day; between = no change. Snapshot cameras: Pillow YCbCr on each JPEG.
Boost	_motion_area_now: the camera's level plus 15 at night, into the same log scale, continued past 100, floor 0.4%.
Readings	Slider units at all times: _motion_level_for_pct takes the boost off. The tuning line adds the mode and colour value.
Clock	Home location from GET /core/api/config through the Supervisor: every 5 min until it answers, then every 12 h. Sunrise and sunset from the US Naval Observatory almanac formula (standard library only), for yesterday, today and tomorrow.

Part	Design
Expectation check	At the first keeper pass after sunset + 1 h (or sunrise + 1 h): a camera watched through the whole window, without a gap over 120 s, that is not in the expected mode is reported once: log warning, cog note, and persistent_notification.create in Home Assistant.
Permission	homeassistant_api: true. The Supervisor's Core proxy refuses only hassio paths (CORE_API_DENY, read in the Supervisor source), so both calls are allowed.

Second round: insects and infrared switches

Evidence: CrystalHeeler's 6 recordings, the add-on log for 21:44 to 07:53, and the 45 recordings in the two camera folders, replayed with OpenCV 4.13.0 on the build PC (it decodes H.265; ffmpeg is not installed).

Finding	Evidence	Change
Insects: one picture	All 12 insect clips: one blurred streak in 1 or 2 pictures (0.14-0.28 s)	Live stream: a 2nd changed picture within 0.5 s, or one of 3% or more (recorded one picture later)
Echoes	Log: one insect as 2-3 equal changes in a row (01:48:17, 1.1% x3). Burst arrival; reference chosen by arrival time	Reference = 4 pictures back in the stream; exact repeats ignored; the insect's own echo 1 s later ignored
Switches	06:20:35: comparisons at 69%, 75%, 31% spread; 06:22:22: a half-switched picture at 19%	2 s hold after a light change; a light change cancels a waiting change
First fix too strict	Requiring a change against both the 1 s and 2 s pictures dropped the cat and 4 porch clips	Not used
Classic view ignored the 30 s cooldown	test system A log, 23:22:38-23:23:07: 6 ffmpeg starts 5 s apart at the Microseven while a 30 s cooldown was in force	Every ffmpeg start waits out the cooldown; a camera in cooldown goes to HTTP snapshots after one failed start
3% rule vs switch	Production replay: 06:20:35 recorded in 4 of 7 (a 24% half-switched picture)	Big change waits one picture for a light change: 0 of 7

Replay through the 2.6.7 code (setting 99, 7 offsets)	Recorded
11 insect clips (ch4 10, ch7 1)	0 of 7 each
1 insect clip (ch4 23:37:28)	1 of 7
2 infrared-colour switches (ch7)	0 of 7 each
Child on a bike, ch7 07:01:53 (log: 4 pictures in a row)	7 of 7
Cat, ch4 03:08:48	4 of 7
Person standing on the porch, ch7 06:59:10	5 of 7
29 other real events	7 of 7 each

Limit: the replay uses the recorded main stream; the live detector watches the sub-stream. The two measured the bike clip differently (replay peak 1.11%, log 1.1-1.2% in 4 pictures in a row), so the field test (A13) decides.

Tests run

Suite	Result	Scope
Release gate	PASS	Seven gates, 45 contracts (8 new, 1 extended for the cooldown fix: the detector format, the boost, the hold, the expectation check, the API call, the location refresh, the motion decision, the file tag), 19 settings in agreement, Dockerfile inputs online.
Python integration	PASS	265 of 265, 61 new. Classic view with a camera in cooldown: one ffmpeg start, then HTTP snapshots. Second round: insect in one picture, with pictures arriving 1, 2 or 3 at a time, and repeated 4 times; a walker; a 6% single picture; a big change then a light change; the 2 s hold; snapshot cameras; name tags. The stand-in detector now varies each picture. Sunrise and sunset for London (June), Sydney (June) and Denver (December) within 4 minutes of published times; polar day gives none. Colour measure on grey and colour JPEGs. The 30 s hold both ways and the dead band. Scale past 100 and the floor. Tuning line and cog payload in slider units. Expectation check: inside the window, missed sunset, missed sunrise, one report only, note cleared, not watched, detector down. Location retry and refresh. The live detector end to end with YUV frames.
JavaScript behaviour	PASS	110 of 110, 4 new: mode line day and night, the note shown and hidden, nothing when disarmed.
Not run	NOTED	ffmpeg is not installed on the build PC (rule 4): the yuv420p frame size (64 x 48 x 1.5 = 4,608 bytes) is from the format's definition, not a run. No real infrared picture was measured: the 2.5 and 5.0 thresholds are estimates. No Docker: the image was not built. The Supervisor calls were tested with a stand-in, not a real Home Assistant.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, Parts 1 to 6, as CLAUDE.md rule 1 requires. Only the rows that 2.6.7's changes touch are scored; the rest carry from the 2.6.6 audit unchanged.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, duplicates, 43 contracts	PASS	207 top-level functions and classes, no duplicates.
1.2 Syntax	compile()	PASS	Gate 1.
1.4 Async/await	Blocking work on the loop	NOTED	Gate 3 clean. The colour measure (1,536 bytes once a second) runs on the loop. The keeper awaits the Home Assistant calls, 10 s timeout each: an unreachable Home Assistant delays one keeper pass every 5 min by up to 10 s.
1.5 Style: line length	Measured added lines	NOTED	45 of 392 added lines exceed 79 characters (up to 100).
1.6 Function design	New helpers	PASS	Ten small functions, each one job: <code>_motion_boost</code> , <code>_motion_area_now</code> , <code>_motion_chroma</code> , <code>_motion_jpeg_chroma</code> , <code>_motion_night_observe</code> , <code>_sun_events_utc</code> , <code>_sun_events_around</code> , <code>_near_sun_event</code> , <code>_night_expectation_check</code> , <code>_ha_api</code> (with <code>_ha_location_refresh</code> and <code>_ha_notify</code>), <code>_motion_decide</code> , <code>_cam_file_tag</code> . <code>_motion_judge</code> now returns a verdict and the percentage.
1.7 Mutability	Shared state	NOTED	Night state joins the global <code>_MOTION</code> dict; <code>_NIGHT_CHECKED</code> is pruned after 2 days. Build plan E2.
1.10 Exceptions	Narrow catches	PASS	<code>aiohttp.ClientError</code> and <code>TimeoutError</code> for Home Assistant calls; <code>OSError</code> , <code>ValueError</code> , <code>DecompressionBombError</code> for JPEG colour.
1.12 Logging	New lines	PASS	Mode switches at INFO with the level change; outside-window switches marked; missed switches at WARNING; colour value in the tuning line.
1.17 Stdlib first	Sun position	PASS	Standard-library math; no astronomy package added.

Standard	Check applied	Result	Note
1.18 Testing	Behavioural tests	NOTED	354 checks, still in the session scratchpad (build plan E7).
2.2 JS async	Cog panel	PASS	No new async code; the mode line fills from the existing poll.
3.3 / 5.1 Escaping	Note text into the page	PASS	Set with textContent, not innerHTML.
5.x CSS in f-string	{{ }} escaping	PASS	Three new rules, braces doubled; page renders (page tests).
6.1 Security	New permission	NOTED	homeassistant_api gives the add-on Home Assistant's REST API. Used for two fixed paths only, no user input in either; token from the environment, never logged.
6.2 Dependencies	None added	PASS	Gate 7.
6.3 Version control	Commits and gate	PASS	One build commit and the release commit; gate run before each; tag after the final run.

Known issues carried forward

- **Colour thresholds are estimates.** Field check in A13 reads the colour value from the tuning line by day and by night.
- **A camera in colour at night under lights** gets no boost, by design; it is reported once a day after sunset.
- **False events at night:** the still-scene peak (0.7%) is above the boosted 100 (0.52%).
- **Fast animals** in view for under 0.5 s and under 3% of the picture are not recorded (the cat: 4 of 7 replays).
- **Several insects at once** can still start a recording.
- Opening the classic view still stops the card's stream (B15).
- Skip Non-Reference Frames breaks H.264 cameras (B3).

Acceptance

- Home Assistant asks for the API permission on update; the log shows 'Home Assistant location received' with sunrise and sunset times.
- After dark, each armed camera logs 'night (IR, black-and-white)' and its cog says 'Night mode (IR): sensitivity +15'.
- A distant walker at night records at a setting that missed it in 2.6.6.
- One hour after sunset, no notification for cameras that switched.
- At dawn, cameras log 'day (colour)' and the setting returns.

- Overnight: far fewer insect recordings; 'changed in one picture only' lines instead; no recording from the dawn or dusk switch.
- Recordings named LorexCH4_<date>_<time>.mp4.

Files changed

File	Change
camera_discovery.py	Night boost, sun clock, Home Assistant calls, cog mode line, light hold, single-picture rejection, stream-clock reference, file tag. Version 2.6.7.
config.yaml	homeassistant_api: true. Version 2.6.7.
verify_release.py	8 new contracts; 4 updated.
docs	Build plan: C15 and B17 done, A13 field test; HTML regenerated.
CHANGELOG.md	2.6.7 entry.